

Sarthak Sahai

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WORK EXPERIENCE

Canadian Space Agency

May 2026 - Present

Student Systems Engineering Intern • Internship

- Processed and analyzed SICK LMS111-10100 LiDAR point clouds in Python, characterizing 1 cm ranging accuracy and a 3 cm minimum detectable obstacle height across a 20 m maximum scan range to support a TRL increase for off-world navigation and landing.
- Defined pushbroom LiDAR surface-mapping configurations for NASA's Artemis 4 by trading 25 Hz and 50 Hz scan rates against 0.25° and 0.5° angular increments across varied pitch angles and traverse speeds.
- Ability to replace a contracted third-party scanning service by building the in-house LiDAR measurement chain end to end - scanning module, Python processing pipeline, and point-cloud viewer

Canada's Wonderland

Apr 2023 - Jan 2024

Rides Mechanical Intern • Internship

- Kept two roller coasters, Dragon Fyre and Leviathan, in safe continuous operation across 8 months through structural integrity assessments, root-cause failure analysis, and preventive mechanical work.
- Diagnosed failures across four restraint and drive subsystems, including lap bars, shoulder restraints, lift chain, and hydraulics, using tolerance analysis, schematics, and manufacturer specifications.

PROJECTS

Night-Optical and Visibility Assessment (NOVA)

Sep 2024 - Aug 2025

Canadian Space Agency (CSA)

- Led a 6-person capstone team as Project and Systems Lead on NOVA, a secondary payload built to a sub-500 g mass allocation and flown on the Canadian Space Agency's RSONar III stratospheric balloon to measure night-sky brightness and support resident-space-object detection at 40 km altitude.
- Owned the systems interface to the RSONar III project manager and the payload's requirements-to-verification traceability, mapping every mechanical, electrical, computational, and environmental requirement to a named test procedure across six subsystems.
- Qualified the Aluminum 6061 structure and 45° camera mount through random-vibration FEM across a 0-2000 Hz PSD profile and a 10 G load case, holding displacement under 0.1 mm and strain within the 8% aluminum limit.
- Brought avionics power draw within the 4 W budget by disabling unused BeagleBone Black Industrial peripherals and retuning exposure rates, holding 3.7 W peak at 5 V over >15 h continuous operation with zero image-data loss.

Reconnaissance Orbital Utility for Terrain Exploration (ROUTE-M)

Jan 2026 - Apr 2026

Space Mission Design, York University

- Served as Systems and Payload Lead for a Mars-orbiting high-definition imaging mission, developed through the NASA project lifecycle to Critical Design Review (CDR).
- Led mission architecture and payload selection for a stereo-imaging system delivering 1 m or better ground sample distance from a 300 km Sun-Synchronous Orbit around Mars.
- Resolved the mission-lifetime versus constellation-size trade by selecting a 2-satellite stereo constellation covering the full Martian surface across a 7-year design life.
- Supported mission-level mass (45 kg payload), power (120 W peak), cost, and risk assessments within a \$750M USD theoretical mission budget.

Ocean Spectroscopy for Carbon and Algal Risk (OSCAR)

Jan 2026 - Apr 2026

Space Mission Design, York University

- Led a 5-person team as overall project lead and payload lead for a space-based ocean spectroscopy mission taken through the NASA project lifecycle to Critical Design Review (CDR)
- Led payload architecture, instrument selection, and interface definition for spectrometer-based measurement of algal formation and ocean health indicators
- Conducted trade studies on spectrometer configurations, orbital considerations, and data quality under mission constraints.

- Resolved the measurement-approach trade by selecting spectroscopic quantification of algal biomass to support ocean-health and global-temperature monitoring goals, delivering a mission design ready to proceed to CDR.

Sparrow - Autonomous Rover Research Platform

Sep 2026 - Present

Canadian Space Agency / York University

- Lead a 9-person team across mechanical, software, and electrical subsystems to build Sparrow, a reconfigurable 1:10-scale wheeled rover test platform for the Canadian Space Agency, targeting a build cost near \$1,500 against commercial platforms such as the Warthog costing many times more.
- Architected a sensor-agnostic interface supporting LiDAR, stereo camera, and monocular camera integration without platform modification, enabling rapid sensor swaps across test campaigns.

Hospital Medicinal Dispensing Robot

Jan 2026 - Apr 2026

Mechatronics, York University

- Led mechanical design and electro-mechanical integration on a 3-person autonomous medicinal dispensing robot, covering component mounting, enclosure design, and interface definition to the ESP32 microcontroller subsystem.
- Delivered the mechanical and electro-mechanical build within a \$300 total project budget.

LEADERSHIP & DESIGN TEAMS

York University Steel Bridge

Sep 2024 - May 2026

President • North York, ON

- Led a 20-member multidisciplinary team through design, fabrication, integration, and testing of a 20 ft competition steel bridge, entered against engineering schools from across Canada at the national competition in Moncton, NB.
- Grew the team from 6 to 20 members and shifted funding from member donations to \$35,000 in external sponsorships and industry collaborations.
- Introduced experimental triangular-truss members in place of the flat members used in traditional builds.

Great Northern Concrete Toboggan Race

Sep 2024 - Feb 2025

Steering and Braking Lead • North York, ON

- Served as Steering and Braking Lead for GNCTR, designing and integrating the toboggan's steering and braking systems in SolidWorks under dynamic load conditions.
- Design was carried over and reused by the following year's team; team won Best New Team and Best Ski Design and was featured in a CP24 news segment.

RESEARCH PROJECT

Observing the HAT-P-18 b Transit and Calculating Planetary Radius through Light Curves

Nov 2024

York University

- Captured 706 CCD frames of exoplanet HAT-P-18 b at 30 s exposures across a single 6-hour transit observation at the Allan I. Carswell Observatory.
- Developed the Python reduction and photometry pipeline using Astropy, CCDPROC, Photutils, and Astroalign to process all 706 frames.
- Computed a planetary radius of 69,112 km from the transit light curve, within 0.6% of the published 69,560 km, alongside transit depth and ingress/egress durations.

SKILLS

- **Simulation & CAD:** Abaqus, FEA, Fusion 360, SAP2000, SolidWorks, STK
- **Hardware & Fabrication:** 3D Printing, DFM/DFA, ESP32, Laser & Water-Jet Cutting, Machine Shop Trained
- **Software & Analysis:** ARM Assembly, Astropy, C++, LiDAR Point-Cloud Processing, MATLAB, Open3D, Python, STK
- **Systems Engineering:** Interface Definition, Requirements Traceability, Trade Studies, Verification & Validation

EDUCATION

Bachelor of Engineering in Space Engineering

York University • North York, ON

Aug 2021 - Apr 2027

Additional Certificates:

- Bergeron Entrepreneurs in Science and Technology (Entrepreneurship and Intellectual Property Law)
- Cross-Disciplinary Mechatronics